

AI and Equity in K-12 Education: An Integrative Analysis Advancing Inclusive and Accountable Design

Abstract

As artificial intelligence (AI) technologies become increasingly integrated into K-12 education, they offer opportunities for instructional innovation while simultaneously raising concerns related to equity, bias, and representation. This study examines how educational AI systems can be designed and implemented in ways that acknowledge and address structural inequities, particularly those affecting historically marginalized learners. While prior research has examined issues of bias, ethics, and instructional design in educational AI, fewer approaches integrate critical social theory with actionable guidance for equitable implementation in K-12 contexts. Grounded in Critical Pedagogy (Freire, 1970), Epistemic Injustice Theory (Fricker, 2007), and Intersectionality (Crenshaw, 1989), the study employed an integrative literature review methodology (Torraco, 2005) to synthesize a purposively selected corpus of 20 peer-reviewed and policy-oriented sources published between 2020 and 2025. Through iterative conceptual synthesis, the review identified recurring equity-related concerns in educational AI, including algorithmic opacity, epistemic exclusion, intersectional harm, and gaps in governance and ethical oversight. These patterns informed the development of the IDEA Framework (Inclusive Design for Equitable AI in Education), a four-domain model encompassing Inclusive Design, Digital Equity, Bias and Justice Analysis, and Ethical Implementation. The framework offers a theoretically grounded and practice-oriented tool for educators, designers, and policymakers to critically evaluate and guide the use of AI systems in K-12 contexts. By integrating equity-focused theoretical perspectives with contemporary scholarship and policy analysis, this study contributes a synthesis-driven framework for more just and accountable AI integration in education.

Keywords: *AI ethics, AI policy, algorithmic bias, educational equity, integrative literature review, K-12 education*

Introduction

As artificial intelligence (AI) tools become increasingly integrated into K-12 classrooms, they are frequently presented as mechanisms for personalization, efficiency, and data-driven instructional improvement (Delcker, 2024; Dwivedi, 2023). Adaptive learning systems, intelligent tutoring platforms, and automated assessment engines promise scalable solutions to longstanding challenges in differentiation and feedback. Yet, as these systems become embedded in everyday instructional practice, they do more than streamline processes. They influence how learning is defined, how performance is evaluated, and how authority is distributed within classrooms. In this study, AI systems refer to a range of data-driven technologies used to support instructional decision-making, including generative AI tools, adaptive learning platforms, and predictive analytics systems. While multiple forms of AI are considered, particular attention is given to generative AI due to its increasing role in classroom writing, feedback, and content creation contexts.

AI technologies do not operate in isolation from social context. They are developed through optimization goals, trained on historically situated datasets, and deployed within educational systems shaped by racial, linguistic, and socioeconomic inequities (Adams, 2021; Mohamed et al., 2020). As a result, the integration of AI into K-12 education raises critical questions: Whose knowledge is treated as normative? Which forms of expression are recognized as legitimate? And how might algorithmic systems stabilize or amplify existing disparities under the appearance of neutrality?

Rather than framing AI bias as a technical malfunction, this study conceptualizes educational AI as a sociotechnical infrastructure that mediates power. Predictive modeling, classification systems, and opaque decision processes can shape instructional pathways,

influence teacher judgment, and affect how students are positioned within academic hierarchies. These dynamics are particularly consequential for students from historically marginalized communities whose linguistic, cultural, and experiential knowledge may not align with dominant algorithmic standards.

Guided by the question, *How can educational AI systems be conceptualized and implemented in ways that identify and challenge embedded biases while advancing equity and justice for diverse learners?*, this study draws on three complementary theoretical perspectives. Critical Pedagogy (Freire, 1970) provides a lens for examining how AI systems shape the student-system relationship and whether algorithmic prescription expands or constrains learner agency. Epistemic Injustice Theory (Fricker, 2007) offers tools for interrogating whose knowledge is granted credibility within AI-mediated environments and how opacity can consolidate epistemic authority. Intersectionality Theory (Crenshaw, 1989) enables analysis of how algorithmic systems may automate and compound structural inequities across overlapping identities.

To explore these concerns, the study employs an integrative literature review (Torraco, 2005), synthesizing interdisciplinary scholarship and policy analyses published between 2020 and 2025. Through theory-informed conceptual integration, recurring patterns related to structural bias, epistemic exclusion, algorithmic opacity, and governance gaps were identified.

These patterns informed the development of the Inclusive Design for Equitable AI in Education (IDEA) Framework, a design-oriented model that translates equity-related concerns into a structured set of considerations for evaluating and guiding AI integration in K-12 contexts.

The framework is developed through integrative synthesis and presented in detail following the analysis.

By connecting critical social theory with contemporary AI scholarship and policy analysis, this study advances examination of how AI systems shape learning environments and educational opportunity. In doing so, it offers both conceptual clarity and practical guidance for more equitable and accountable AI integration in schools.

Theoretical Foundations and Analytic Orientation

This study is grounded in three complementary theoretical traditions: Critical Pedagogy (Freire, 1970), Epistemic Injustice Theory (Fricker, 2007), and Intersectionality Theory (Crenshaw, 1989). Rather than functioning as parallel philosophical backdrops, these frameworks serve as analytic lenses that guide the interpretation of contemporary scholarship on artificial intelligence (AI) in K-12 education. Combined, they enable examination of how AI systems mediate power at pedagogical, epistemic, and structural levels.

Each framework provides a distinct analytic contribution. Critical Pedagogy directs attention to the student-system relationship and the regulation of learner agency within AI-mediated environments (Farag et al., 2021; Freire, 1970). Epistemic Injustice Theory interrogates the legitimacy and distribution of knowledge authority embedded in algorithmic systems, particularly in contexts characterized by opacity and automated classification (Fricker, 2007; Kay et al., 2024; Van Slyke et al., 2025). Intersectionality Theory highlights how AI systems may automate and scale historical inequalities through proxy variables, data patterns, and

institutional decision structures that disproportionately affect students situated at overlapping margins of identity (Crenshaw, 1989; Mohamed et al., 2020; Rice et al., 2019).

As a whole, these perspectives allow the study to move beyond identifying bias as a technical flaw and instead examine how algorithmic systems can structurally produce and stabilize inequities within educational contexts (Adams, 2021; Gaskins, 2022; Mohamed et al., 2020).

Critical Pedagogy: Agency and Algorithmic Prescription

Critical Pedagogy, rooted in Freire's (1970) critique of hierarchical and transmission-based models of education, positions learning as a dialogic process grounded in agency, reflection, and co-construction of knowledge. Within AI-mediated classrooms, this lens raises questions about whether algorithmic systems expand or constrain student participation in meaning-making (Farag et al., 2021).

Many contemporary AI tools rely on predictive modeling and optimization goals that prescribe instructional pathways based on prior performance data (Dwivedi, 2023; Mohamed et al., 2020). While such systems are frequently framed as mechanisms for personalization, scholars caution that optimization criteria inevitably encode normative assumptions about what counts as progress, proficiency, or risk. When adaptive platforms determine pacing, sequencing, or feedback through automated classification, they risk narrowing classroom dialogue. Building on critical scholarship that interrogates optimization-driven educational systems (Adams, 2021; Gaskins, 2022; Mohamed et al., 2020), this study conceptualizes this dynamic as *algorithmic*

prescription: a unidirectional configuration of learning shaped by data optimization rather than negotiated meaning-making.

Building on critical scholarship that interrogates the automation of pedagogical judgment (Adams, 2021; Mohamed et al., 2020; Noble, 2018), this study uses Critical Pedagogy to examine whether AI systems function as dialogic supports that amplify student voice and teacher agency or whether they introduce forms of pedagogical determinism under the appearance of efficiency.

Epistemic Injustice: Knowledge Authority and Algorithmic Opacity

Epistemic Injustice Theory (Fricker, 2007) provides tools for analyzing how credibility, legitimacy, and interpretive authority are distributed within knowledge systems. In AI-mediated environments, these concerns surface in at least two ways: testimonial exclusion and hermeneutical opacity (Kay et al., 2024; Sardelli, 2022).

AI systems are trained on datasets that reflect dominant linguistic, cultural, and behavioral norms (Cave & Dihal, 2020; Mohamed et al., 2020; Noble, 2018). When such norms become embedded as algorithmic “ground truths,” student expressions that diverge from those norms may be labeled as error, deficiency, or risk indicator rather than legitimate variation. This dynamic reflects testimonial injustice, wherein certain forms of knowledge or expression are systematically granted less credibility (Fricker, 2007; Kay et al., 2024). In educational settings, this may manifest through automated scoring systems that misinterpret dialectal variation,

predictive tools that flag students based on historically biased disciplinary data, or language models that privilege standardized linguistic forms (Gaskins, 2022; Sardelli, 2022).

Hermeneutical injustice emerges when students and educators lack access to the interpretive logic behind algorithmic decisions (Fricker, 2007; Karasinski & Candiott, 2024). The “black box” problem represents an epistemic consolidation of authority within the system itself, not simply a transparency challenge (Karasinski & Candiott, 2024; Long & Magerko, 2020). When algorithmic reasoning is inaccessible or unintelligible, students are deprived of the ability to understand or contest the logic governing their assessment, placement, or classification (Kay et al., 2024; Van Slyke et al., 2025). Through this framework, the study interrogates not only whether AI systems are accurate, but whether they are epistemically accountable and pedagogically intelligible.

Intersectionality: Structural Magnitude and Automated Hierarchy

Intersectionality Theory (Crenshaw, 1989) foregrounds how overlapping systems of power, such as race, gender, class, language status, and ability, interact to produce compounded forms of marginalization. Applied to educational AI, this lens highlights how algorithmic systems may automate and scale structural inequalities rather than merely reflect them (Eubanks, 2018; Mohamed et al., 2020).

Many AI tools rely on proxy variables, such as zip codes, attendance histories, disciplinary records, or standardized performance metrics, that correlate with broader structural inequities (Eubanks, 2018; Selbst et al., 2019). When these proxies are incorporated into predictive models or classification systems, they can reproduce historical hierarchies through automated decision processes that appear neutral but reflect unequal institutional conditions

(Gaskins, 2022; Mohamed et al., 2020). Students situated at the intersections of linguistic marginalization, racial minoritization, and socioeconomic disadvantage may therefore experience intensified forms of algorithmic misclassification or restricted opportunity (Dunne, 2022; Rice et al., 2019).

This perspective shifts analysis from isolated bias to structural patterning. It invites examination of how AI systems participate in what may be understood as automated hierarchy: the scaling of historically embedded inequalities through computational processes embedded in institutional infrastructures.

Integrative Analytic Orientation

These three theoretical foundations operate in concert throughout the integrative review. Critical Pedagogy identifies the pedagogical location of harm in AI-mediated student-system interactions (Farag et al., 2021; Freire, 1970). Epistemic Injustice Theory clarifies the mechanism of harm through credibility deficits, opacity, and labeling authority (Fricker, 2007; Kay et al., 2024). Intersectionality Theory reveals the structural magnitude of harm as algorithmic systems compound existing inequities across intersecting identities (Crenshaw, 1989; Mohamed et al., 2020).

By employing these frameworks as analytic lenses rather than descriptive backdrops, the study synthesizes interdisciplinary scholarship to examine how AI systems shape agency, authority, and opportunity in K-12 education. This theoretical orientation directly informs the integrative review process and guides the conceptual development of the Inclusive Design for Equitable AI in Education (IDEA) Framework presented later in the paper.

Methodology: Integrative Literature Review

Research Design

This study employed an integrative literature review methodology (Torraco, 2005) to synthesize a purposively selected corpus of 20 peer-reviewed and policy-oriented sources published between 2020 and 2025. An integrative review was selected because the intersection of AI ethics, critical theory, and K-12 educational practice remains an emerging and interdisciplinary field. Rather than aggregating effect sizes or seeking exhaustive empirical coverage, the purpose of this review was conceptual integration and theory-building, specifically, the development of the Inclusive Design for Equitable AI in Education (IDEA) Framework.

Sources were selected to support depth of conceptual synthesis rather than comprehensive enumeration. Inclusion criteria required that sources (a) examine artificial intelligence in K-12 educational contexts, (b) engage questions of equity, bias, ethics, governance, or sociotechnical design, and (c) provide either empirical findings, conceptual argumentation, or policy analysis relevant to educational AI. Exclusion criteria eliminated studies focused exclusively on higher education, purely technical machine-learning optimization without pedagogical framing, or non-scholarly opinion pieces.

Searches were conducted across Google Scholar, ERIC, JSTOR, and EBSCOhost databases using combinations of the following terms: ("Artificial Intelligence" OR "AI") AND ("K-12" OR "school") AND ("equity" OR "bias" OR "justice" OR "ethics"). Snowball sampling from foundational and frequently cited works was used to identify additional relevant scholarship. This multi-stage process yielded a bounded yet conceptually rich corpus.

The analytic process involved iterative comparison across sources to identify recurring patterns, tensions, and conceptual relationships. Analysis proceeded through a hybrid deductive-inductive approach. Deductive coding was informed by constructs derived from Critical Pedagogy (e.g., agency, dialogic learning), Epistemic Injustice Theory (e.g., testimonial authority, hermeneutical opacity), and Intersectionality Theory (e.g., compounded marginalization, structural power). Inductive coding was simultaneously employed to identify recurring technical and policy manifestations in the literature, such as optimization bias, proxy-variable reliance, surveillance mechanisms, and algorithmic opacity.

Constant comparison and analytic memoing supported reflexive engagement with the corpus, allowing themes to be refined and integrated across disciplinary boundaries. Coding and memoing were conducted through systematic highlighting and analytic annotations within a structured review document to ensure traceability of thematic development across sources. To enhance analytic transparency, coding decisions were documented through iterative memoing and maintained within a structured audit trail that traced the development of themes across sources. This process supported reflexive engagement with the data and ensured that thematic interpretations remained grounded in the literature rather than isolated analytic judgments. Recurring conceptual domains were then synthesized into higher-order thematic groupings, which informed the development of the IDEA Framework. The framework therefore reflects patterns observed across the integrated literature rather than any single theoretical or empirical source.

Source Selection and Corpus Development

Source selection followed a multi-stage, theory-informed process designed to ensure conceptual relevance and disciplinary breadth. Building on the inclusion and exclusion criteria described above, sources were evaluated for their substantive engagement with equity-related concerns in K-12 educational AI, including structural bias, epistemic exclusion, algorithmic opacity, governance gaps, and classroom-level implications.

Initial searches were conducted using the Boolean strings described in the Research Design section across Google Scholar, ERIC, JSTOR, and EBSCOhost databases. Targeted keyword searches included phrases such as “AI bias in education,” “algorithmic injustice,” “K-12 surveillance,” and “equity in educational technology.” Titles and abstracts were screened for alignment with the study’s theoretical orientation, followed by full-text review of selected articles.

To expand disciplinary coverage, reference lists from foundational and frequently cited works were examined to identify additional relevant scholarship. This combination of purposive sampling and snowball techniques yielded a final corpus of 20 interdisciplinary sources, including peer-reviewed journal articles, scholarly books, policy reports, and published analyses of educational AI tools.

The resulting corpus was intentionally bounded yet conceptually rich, prioritizing analytic depth and theoretical coherence over numerical breadth. Because the aim of the review was integrative synthesis rather than empirical generalization, the corpus was curated to support cross-theoretical dialogue and framework development. A complete list of analyzed sources is provided in Appendix A.

Analytic and Synthesis Process

The literature corpus functioned as the primary analytic data for this integrative review. Each source was read in full and examined through iterative, theory-informed comparison to identify recurring patterns, tensions, and conceptual relationships related to equity in educational AI systems.

Analysis proceeded through a hybrid deductive-inductive process. Deductively, analytic attention was guided by constructs derived from Critical Pedagogy (Freire, 1970), Epistemic Injustice Theory (Fricker, 2007), and Intersectionality Theory (Crenshaw, 1989). These frameworks informed initial analytic questions, including:

- How does the AI system shape or regulate learner agency?
- Whose knowledge is granted credibility or marginalized?
- How might algorithmic processes compound inequities across intersecting identities?

These guiding questions functioned as interpretive lenses rather than rigid coding templates, allowing theoretical constructs to orient the reading of each text while preserving openness to emergent patterns.

Simultaneously, inductive analysis was used to identify recurring technical and policy-level manifestations within the literature, such as optimization bias, proxy-variable reliance, algorithmic opacity, surveillance mechanisms, and governance gaps. These patterns were documented through systematic highlighting and analytic memoing, enabling traceability of conceptual development across sources.

Constant comparison was employed throughout the review process to examine similarities and differences in how AI systems were described, justified, critiqued, and

contextualized across disciplinary traditions. Analytic memos captured emerging conceptual linkages, areas of convergence and divergence, and theoretical tensions.

Through iterative synthesis, recurring themes were grouped into higher-order conceptual domains. These domains reflected not only documented equity concerns but also the sociotechnical mechanisms through which such concerns were produced and stabilized. The resulting conceptual structure directly informed the development of the IDEA Framework, ensuring that the framework emerged from patterned synthesis rather than from a priori categorization.

Conceptual Integration and Framework Development

Through iterative comparison and analytic memoing, recurring equity-related patterns were identified across the literature corpus. Rather than emerging from predetermined theoretical categories, these patterns reflected convergences in how scholars described the sociotechnical mechanisms, institutional conditions, and educational consequences associated with AI implementation in K-12 settings. Over successive readings, patterns were refined, compared, and conceptually grouped into higher-order thematic domains.

While the analytic orientation was informed by Critical Pedagogy (Freire, 1970), Epistemic Injustice Theory (Fricker, 2007), and Intersectionality Theory (Crenshaw, 1989), the resulting domains were not deductively mapped onto these frameworks. Instead, the theories guided interpretation of recurring concerns, helping to clarify the pedagogical, epistemic, and structural dimensions of observed patterns. Domain construction reflected convergence across interdisciplinary scholarship rather than a priori theoretical alignment.

Through this integrative process, four interconnected conceptual domains were identified:

1. Epistemic and Social Justice
2. Structural and Design Injustice
3. Ideological and Historical Foundations
4. Policy and Ethical Oversight

Each domain represents a distinct yet interrelated dimension of how AI systems mediate power within educational contexts.

Epistemic and Social Justice captures recurring concerns about whose knowledge is recognized as legitimate within AI-mediated systems and how credibility is distributed through algorithmic processes. Across the corpus, scholars examined how linguistic norms, cultural assumptions, and standardized benchmarks become encoded into AI training data and assessment models, shaping what counts as valid expression or demonstrated understanding (Cave & Dihal, 2020; Kay et al., 2024; Sardelli, 2022). These dynamics produce credibility differentials in automated scoring, predictive classification, and feedback systems, particularly for students whose dialects, epistemologies, or modes of expression diverge from dominant norms. This domain highlights how inequity may emerge not only from inaccurate outputs but from systemic patterns of epistemic marginalization embedded in AI-mediated evaluation.

Structural and Design Injustice reflects patterns related to training data composition, proxy-variable reliance, accessibility limitations, and the embedding of institutional bias within optimization processes. The literature repeatedly emphasized that AI systems are constructed within unequal social infrastructures and frequently rely on variables, such as disciplinary records, attendance histories, or demographic proxies, that correlate with broader structural

inequities (Eubanks, 2018; Mohamed et al., 2020; Selbst et al., 2019). These design choices can automate and scale historical disparities through predictive modeling and classification systems that appear neutral but operate within stratified institutional conditions. This domain therefore foregrounds inequity at the level of architecture, emphasizing how design decisions shape opportunity structures before tools are ever implemented in classrooms.

Ideological and Historical Foundations emerged from scholarship interrogating the normative assumptions embedded in AI development and adoption. Across the corpus, authors challenged narratives of technological neutrality, objectivity, and inevitability that frequently accompany AI innovation in education (Adams, 2021; Mohamed et al., 2020; Noble, 2018). These narratives position AI outputs as inherently authoritative, obscuring the value systems embedded in data selection, optimization criteria, and performance metrics. Historical analyses further traced how contemporary AI infrastructures are shaped by longer-standing patterns of bureaucratic standardization, racialized classification, and data colonialism (Eubanks, 2018; Noble, 2018). This domain therefore captures the ideological logics that normalize AI systems as objective tools while masking the historical power relations that inform their construction.

Policy and Ethical Oversight encompasses concerns regarding governance infrastructure, transparency requirements, accountability mechanisms, and institutional responsibility in AI adoption. Sources consistently emphasized that fragmented or absent policy guidance increases the likelihood that AI systems will be implemented without clear standards for explainability, bias monitoring, or stakeholder participation (Eutsler et al., 2025; UNESCO, 2022; U.S. Department of Education, 2023). Without structured oversight, algorithmic decision-making may become normalized within routine educational processes, including assessment, placement, and

behavioral monitoring. This domain highlights how governance structures mediate the extent to which AI systems are interrogated, regulated, or left to operate with limited public scrutiny.

These four domains represent analytically distinct but mutually reinforcing dimensions through which AI systems mediate power in educational settings. Together, they demonstrate that AI operates simultaneously as a mechanism of knowledge validation, a structural design system, an ideological project, and a policy-regulated infrastructure.

The Inclusive Design for Equitable AI in Education (IDEA) Framework was developed as an applied response to these synthesized domains. While the domains articulate recurring equity tensions documented across scholarship, the IDEA Framework translates these insights into design-oriented analytic categories intended to guide evaluation, implementation, and policy decision-making in K-12 contexts. In this way, the framework operationalizes conceptual synthesis into actionable structure while preserving the complexity of the sociotechnical conditions from which it emerged. Figure 1 illustrates the relationship between the conceptual domains and the applied domains of the IDEA Framework, highlighting how theoretical synthesis informs practical guidance without collapsing one into the other.

Integrative Synthesis of Equity Concerns in Educational AI

Structural Bias and Systemic Inequities in Educational AI

Artificial intelligence systems in K-12 education are frequently positioned as mechanisms for personalization, efficiency, and data-informed instructional decision-making (Akgun, 2021; Delcker, 2024; Dwivedi, 2023). However, the integrative synthesis reveals that these systems operate within, and are shaped by, existing structural inequalities embedded in educational and

technological infrastructures. Rather than functioning as neutral instructional supports, AI tools are developed through design choices, data practices, and optimization logics that can reproduce and intensify systemic inequities.

Across the literature, scholars emphasize that structural bias in educational AI does not arise solely from isolated technical errors but from the sociotechnical conditions under which systems are constructed (Adams, 2021; Mohamed, et al., 2020). Training datasets, labeling decisions, outcome optimization criteria, and development team composition influence whose knowledge is encoded as normative and whose experiences are marginalized. As a result, inequity may be embedded in AI systems prior to classroom implementation, shaping how students are classified, evaluated, and supported.

A primary mechanism through which structural bias emerges is the composition of training datasets and development teams. Adams (2021) and Mohamed, et al. (2020) document how facial recognition and natural language processing tools consistently underperform for racially minoritized students due to the overrepresentation of white, Western users in foundational datasets. In educational contexts, such discrepancies can translate into misidentification, skewed performance evaluations, or disproportionate disciplinary interventions. These outcomes reflect not merely technical limitations but systemic patterns of exclusion embedded in data selection and model training processes. Kamran (2024) and Muldoon (2023) further argue that these patterns are indicative of limited cultural responsiveness in AI development, where the lived experiences of marginalized communities are insufficiently represented in system design.

Linguistic bias represents another structural dimension of inequity in educational AI. Large language models and automated assessment systems frequently privilege dominant language norms while misinterpreting non-dominant dialects, multilingual speech patterns, and culturally rooted communication styles (Cave, 2020; Kohnke, 2024). In classroom settings, such systems may penalize linguistic variation by equating deviation from standardized norms with error or deficiency (Sardelli, 2022; Van Slyke, 2025). These patterns reflect what Fricker (2007) conceptualizes as epistemic injustice, wherein credibility deficits are imposed upon speakers whose knowledge practices fall outside dominant interpretive frameworks. When automated tools fail to recognize diverse linguistic expressions as legitimate, they risk reinforcing hierarchies of knowledge that marginalize already underrepresented students.

Gender- and ability-based inequities further illustrate how design assumptions encode exclusionary norms. Gross (2023) observes that default avatar representations and system voice interfaces often reinforce gender stereotypes, shaping students' perceptions of authority and belonging. Such representations rarely account for nonbinary or gender-diverse learners. Similarly, accessibility limitations, such as speech-recognition systems that struggle with atypical speech patterns or platforms that fail to accommodate cognitive and motor differences, demonstrate how ableist assumptions can become embedded in system architecture when disability is not intentionally considered during development (Hutchinson et al., 2020; Schwartz, 2022). These examples underscore how AI systems may inadvertently privilege dominant identities while constraining inclusive participation.

Intersectionality further clarifies that these biases do not operate independently but compound for learners situated at overlapping margins of identity (Crenshaw, 1989). Students who are simultaneously linguistically minoritized, racially marginalized, and socioeconomically

disadvantaged may experience algorithmic harms that exceed single-axis analyses (Dunne, 2023; Rice et al., 2019). Predictive systems that rely on proxy variables such as zip code, disciplinary history, or language status can disproportionately misclassify such students, narrowing access to advanced coursework or supportive interventions. In this way, AI systems may automate and scale structural hierarchies that are already present in educational institutions.

This thematic synthesis underscores that educational AI systems function as sociotechnical infrastructures that both reflect and reproduce institutional power dynamics. Structural bias arises from data practices, design assumptions, and optimization logics embedded in broader social hierarchies. Without deliberate intervention, these systems risk stabilizing inequities under the appearance of technological objectivity. From a critical pedagogical perspective, such dynamics may reposition educators and students within technocratic structures in which algorithmic outputs assume implicit authority, narrowing opportunities for dialogic judgment and reflective engagement (Freire, 1970). Addressing structural bias therefore requires more than technical refinement; it demands critical examination of whose knowledge is represented, whose experiences are prioritized, and how institutional values are encoded in system design.

Algorithmic Opacity and Accountability

As AI systems become increasingly embedded in K-12 classrooms, concerns about transparency and explainability have gained prominence across the literature. Many educational AI tools rely on complex machine learning models that generate outputs without providing intelligible explanations for how those outcomes were reached, a phenomenon commonly described as the “black box” problem (Karasinski, 2024). In pedagogical contexts, this opacity

limits educators' ability to interpret, question, or contextualize AI-generated feedback, particularly when such outputs inform grading decisions, behavioral assessments, or instructional pathways.

The synthesis indicates that opacity presents both epistemic and professional challenges. When teachers cannot understand how or why an AI system produced a recommendation, they may either distrust the system entirely or accept its outputs without sufficient scrutiny under the assumption of technical objectivity. As Long and Magerko (2020) explain, the absence of explainable reasoning constrains educators' capacity to challenge flawed recommendations, thereby redistributing epistemic authority from classroom professionals to algorithmic systems. From the perspective of epistemic injustice (Fricker, 2007), such opacity can create hermeneutical gaps in which students and teachers lack the interpretive resources necessary to understand or contest the criteria governing evaluation. In classrooms shaped by developmental, linguistic, and cultural variation, this displacement of interpretive authority risks marginalizing teacher expertise and undermining trust among students and families (Dwivedi, 2023; Nazaretsky, 2022).

Efforts to address these concerns have generated increased interest in explainable AI, systems designed to offer human-understandable justifications for their outputs; however, the literature cautions that explainability must be pedagogically meaningful rather than merely technically available. As Dwivedi (2023) notes, an explanation sufficient for a developer may remain inaccessible to a classroom teacher or irrelevant to a student's lived context. For explainability to support equitable instruction, it must be designed with attention to user expertise, contextual variability, and classroom realities. Without such grounding, transparency mechanisms risk functioning as procedural formalities rather than substantive safeguards.

Closely related patterns across sources involve issues of rigid classification and assessment bias. Many AI systems used in education rely on predictive models trained on standardized benchmarks that may not reflect the diverse ways students demonstrate knowledge or progress. Cave (2020), Gaskins (2022), and Sardelli (2022) document how such systems can misclassify student responses based on accent, dialect, or culturally unfamiliar references, particularly within automated scoring engines and behavioral prediction tools. These misclassifications do not necessarily signal a lack of understanding but often reveal misalignment between students' expressive modes and the system's trained parameters.

The consequences of such misalignments extend beyond isolated technical inaccuracies. Inaccurate classifications may influence placement in remedial programs, restrict access to enrichment opportunities, or contribute to unwarranted disciplinary actions. Muldoon (2023) and Van Slyke (2025) emphasize that these harms stem from the disconnect between centralized algorithmic design and the localized, relational nature of classroom teaching. Opaque and rigidly standardized decision processes can subordinate professional judgment to automated outputs. From a critical pedagogical standpoint (Freire, 1970), such dynamics risk narrowing dialogic engagement and reinforcing technocratic governance of learning.

Collectively, this synthesis illustrates that opacity and rigid classification criteria are not merely technical design challenges but matters of epistemic legitimacy and institutional accountability. Equitable AI integration therefore hinges not solely on model performance but on governance structures that render algorithmic processes transparent, open to challenge, and accountable to the educational communities they affect. Without such accountability, AI systems may consolidate authority in ways that obscure responsibility and diminish professional agency within educational environments.

Toward More Equitable and Ethical AI Integration

While the literature consistently documents concerns regarding algorithmic bias and opacity in educational AI (Gaskins, 2022; Karasinski, 2024; Mohamed, et al., 2020), it also articulates a growing set of responses aimed at guiding more equitable and ethically grounded integration (Delcker, 2024; Kamran, 2024; Schwartz, 2022). Rather than positioning AI harms as inevitable byproducts of innovation, scholars increasingly frame them as governance and design challenges that can be addressed through intentional policy infrastructure, participatory development, and sustained accountability mechanisms (Dwivedi, 2023; Mohamed, et al., 2020; Muldoon, 2023).

A recurring theme across the literature is the absence of consistent policy frameworks to guide ethical AI implementation in K-12 education. Despite expanding adoption, many school districts operate without formal guidelines addressing bias mitigation, data governance, transparency, or accountability (Eutsler et al., 2025; Kohnke, 2024). In response, scholars and practitioners have developed educator-facing toolkits and structured policy guidance, such as Kohnke's AI Toolkit and Eutsler et al.'s (2025) seven-step framework for district-level AI policy development. These efforts align with broader regulatory models, including the European Union's AI Act and UNESCO's (2022) Recommendation on the Ethics of Artificial Intelligence, which emphasize human-centered and rights-based governance approaches. However, translating high-level policy principles into actionable district-level practices remains challenging within decentralized educational systems, particularly where technical expertise and evaluative capacity are limited.

Beyond formal regulation, the literature emphasizes participatory design as a structural strategy for mitigating inequity. Rather than treating educators and students as passive recipients of technology, co-design approaches position them as collaborators in shaping system functionality and instructional goals (Delcker, 2024; Schwartz, 2022). Such models redistribute decision-making authority and increase contextual responsiveness, thereby addressing both epistemic and structural dimensions of bias identified in earlier sections. When development processes incorporate classroom insight, cultural knowledge, and professional expertise, AI tools are more likely to align with diverse learner needs.

Inclusive design is similarly foregrounded as foundational to equity-oriented AI development. Gaskins (2022), Kamran (2024), and Muldoon (2023) caution that homogenous development teams and narrow training datasets risk reproducing dominant cultural assumptions while overlooking marginalized perspectives. In contrast, inclusive design prioritizes representational diversity in both data and decision-making processes. As Mohamed, et al. (2020) argues, such approaches require reexamining whose experiences are treated as normative and who holds epistemic authority in defining the purposes of AI in education. This shift challenges assumptions of universality and supports systems adaptable to varied linguistic, cultural, and ability-based contexts.

Scholars further emphasize that bias detection and monitoring should function as continuous processes rather than one-time technical adjustments. Dwivedi (2023) and Mohamed, et al. (2020) advocate for ongoing impact audits, stakeholder feedback mechanisms, and iterative system refinement responsive to evolving classroom conditions. In K-12 environments, this may involve systematically examining whether AI systems disproportionately misclassify students

across race, language, gender, or ability categories and implementing corrective adjustments informed by educator input.

These proposals reflect a broader reorientation of educational AI toward governance, inclusion, and shared responsibility. Ethical integration depends not solely on technological advancement but on sustained institutional commitments to transparency, accountability, and participatory oversight. By foregrounding collaboration among developers, educators, policymakers, and communities, the literature outlines pathways for aligning AI systems with educational values and social justice commitments rather than allowing efficiency logics to dominate instructional decision-making.

Translating Conceptual Domains into the IDEA Framework

Building on the thematic and theoretical synthesis presented in the preceding sections, the literature was further integrated to identify higher-order conceptual domains that capture the central equity-related tensions associated with artificial intelligence in K-12 education. These domains did not arise from predetermined categories but from recurring patterns of convergence across recent scholarship, reflecting sustained concerns related to epistemic legitimacy, structural design, ideological framing, and governance infrastructure in educational AI systems.

Through this integrative process, four interconnected conceptual domains were identified: Epistemic and Social Justice, Structural and Design Injustice, Ideological and Historical Foundations, and Policy and Ethical Oversight. Together, these domains represent distinct yet interrelated dimensions through which inequities in educational AI are produced, normalized, and potentially mitigated. Each domain synthesizes multiple strands of interdisciplinary

scholarship, highlighting how sociotechnical systems intersect with educational values, institutional practices, and broader systems of power.

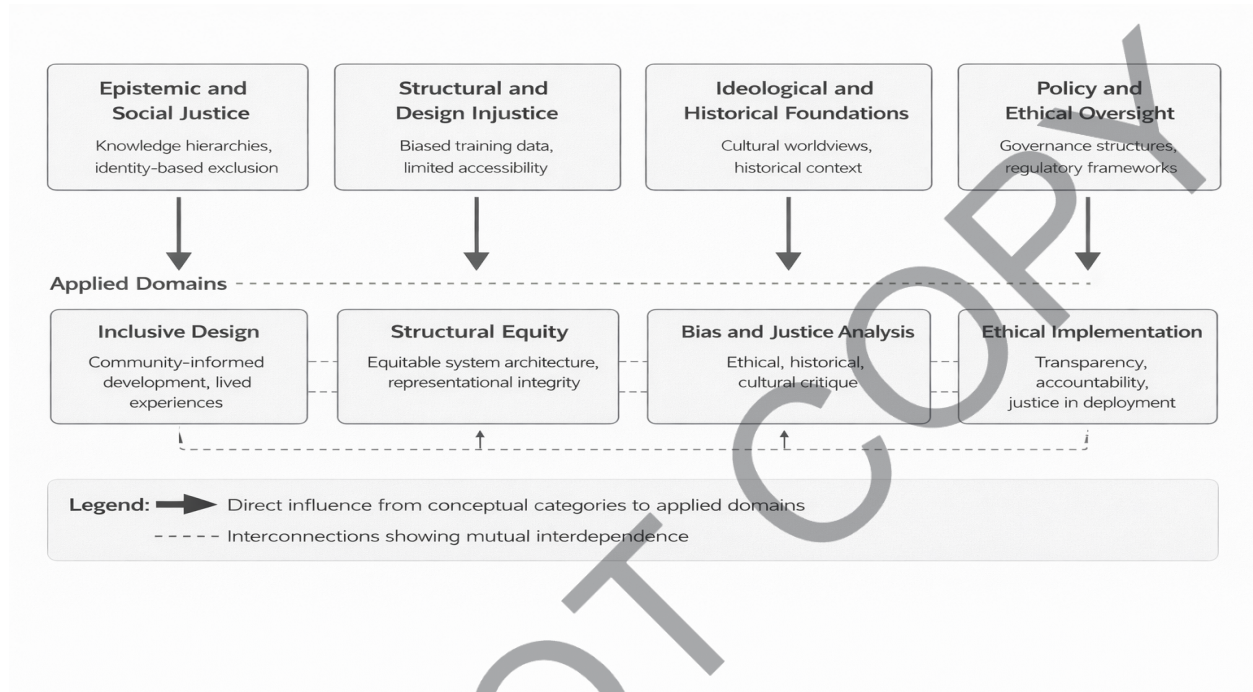
These conceptual domains serve as the analytic foundation for the Inclusive Design for Equitable AI in Education (IDEA) Framework. In the following section, the IDEA Framework is introduced as a design-oriented model that translates these synthesized insights into applied categories intended to guide evaluation, implementation, and policy development in K-12 contexts. In this way, the framework operationalizes conceptual analysis into actionable structure while preserving the complexity of the underlying equity concerns.

Inclusive Design for Equitable AI in Education (IDEA) Framework Development

The Inclusive Design for Equitable AI in Education (IDEA) Framework emerged through integrative synthesis of contemporary scholarship examining equity, power, and ethical responsibility in educational artificial intelligence. Drawing on recurring patterns identified across peer-reviewed and policy-oriented sources published between 2020–2025, the framework represents an authorial integration of documented concerns related to epistemic exclusion, structural inequity, ideological framing, and governance gaps in K-12 educational AI. Rather than serving as a checklist, the IDEA Framework functions as a design-oriented analytic guide for evaluating and shaping AI systems in educational contexts (Delcker, 2024; Dwivedi, 2023; Gaskins, 2022). Figure 1 illustrates the relationship between the foundational conceptual domains and the applied domains of the IDEA Framework. The IDEA Framework is not intended to function as a linear or checklist-based model, but rather as a set of interrelated lenses that can be applied flexibly depending on context. Practitioners may engage with one or multiple

domains simultaneously, using the framework to guide reflective interpretation and decision-making rather than to produce uniform or prescriptive outcomes.

Figure 1: IDEA Framework: Conceptual Categories and Applied Domains



The framework is organized around four applied domains: Inclusive Design, Structural Equity, Bias and Justice Analysis, and Ethical Implementation. These domains correspond to four foundational conceptual areas synthesized across the literature: Epistemic and Social Justice, Structural and Design Injustice, Ideological and Historical Foundations, and Policy and Ethical Oversight. Together, these conceptual foundations illuminate how AI systems mediate power, knowledge, identity, and governance in educational settings, while the applied domains translate those insights into actionable considerations for design, evaluation, and implementation (Eutsler et al., 2025; Kohnke, 2024; Muldoon, 2023).

The IDEA Framework is intended to function as a structured interpretive and decision-making tool for educators, instructional designers, and policymakers. Rather than prescribing

specific technologies or practices, it provides a set of guiding considerations that can be applied when selecting, implementing, or evaluating AI systems in educational contexts. In this way, the framework supports reflective judgment by prompting users to examine how AI tools shape participation, access, legitimacy, and ethical responsibility across instructional settings.

The Epistemic and Social Justice foundation responds to concerns regarding whose knowledge is recognized and legitimized within AI-mediated educational systems. Studies of language-based AI tools and automated assessment systems demonstrate how dominant linguistic norms and standardized representations of knowledge are treated as neutral baselines, while dialectal variation, multilingual expression, and community-based epistemologies are rendered invisible or deficient (Helm et al., 2023; Sardelli, 2022). As Kay (2024) argues, such systems do not simply misinterpret student expression but actively reshape what counts as valid knowledge within algorithmically structured learning environments. Within the IDEA Framework, this foundation informs the Inclusive Design domain, which emphasizes participatory, community-informed development practices that position learners and educators as epistemic contributors rather than passive users (Delcker, 2024; Kohnke, 2024). For example, in a secondary English classroom using a generative AI writing assistant, educators might examine whether the system recognizes and supports diverse linguistic patterns or privileges standardized academic English as the default. Applying an inclusive design lens would involve evaluating how student voice is represented in AI-generated suggestions and whether the tool allows for meaningful revision that preserves individual expression. In this way, inclusive design extends beyond access to participation in knowledge construction.

The Structural and Design Injustice foundation reflects how inequities become embedded in AI systems through decisions related to training data, accessibility, functionality, and user

representation. Contemporary research documents how data-driven systems can automate and scale existing inequalities when design processes fail to account for structural disadvantage or exclude affected communities from decision-making (Gaskins, 2022; Mohamed, et al., 2020). Muldoon (2023) notes that such inequities are often framed as technical limitations rather than structural design choices. This foundation grounds the Structural Equity domain of the IDEA Framework, which focuses on equitable system architecture, representational integrity, and the infrastructural conditions necessary for AI systems to expand rather than restrict educational opportunity. In practice, structural equity concerns emerge in how generative AI systems are trained and configured to represent learners and learning processes. For example, an AI-powered writing or tutoring system trained predominantly on standardized academic texts may fail to reflect the linguistic diversity and rhetorical practices of the students it serves. This can result in feedback or recommendations that subtly reshape student expression toward dominant norms. A structural equity perspective would prompt educators and designers to examine how training data, system assumptions, and representational choices influence whose knowledge is reflected and whose is marginalized within AI-mediated learning environments.

The Ideological and Historical Foundations domain situates educational AI within broader intellectual and cultural traditions that shape how technology is framed as objective, neutral, or inevitable. Recent scholarship critiques the dominance of technocratic and positivist assumptions in AI development, arguing that these perspectives obscure the normative values embedded in algorithmic systems and limit attention to historical and cultural context (Dwivedi, 2023; Mohamed et al., 2020). Without explicit interrogation of these assumptions, AI tools risk reinforcing existing hierarchies under the guise of innovation. This foundation informs the Bias and Justice Analysis domain of the IDEA Framework, which encourages ethical, historical, and

cultural critique of AI tools to surface hidden assumptions, exclusions, and power relations that may otherwise remain unexamined. For instance, when generative AI systems are used to provide feedback on student writing, educators might examine how the system interprets linguistic variation, tone, or rhetorical style. If feedback consistently redirects students toward dominant discourse patterns, this may reflect embedded bias in training data or evaluation criteria. A bias and justice analysis would prompt educators to question whose language practices are treated as normative and how algorithmic judgments shape perceptions of legitimacy and competence.

Finally, the Policy and Ethical Oversight foundation addresses the governance structures and institutional commitments that shape how AI is adopted and regulated in educational contexts. Studies highlight that fragmented or absent policy guidance increases the likelihood that AI systems will operate without sufficient transparency, accountability, or public scrutiny, particularly in K-12 environments (Eutsler et al., 2025; Quinn, 2024). Van Slyke (2025) cautions that without clear oversight mechanisms, educational AI may normalize surveillance and automated decision-making practices that disproportionately affect marginalized students. This foundation underpins the Ethical Implementation domain of the IDEA Framework, which foregrounds transparency, accountability, and ongoing evaluation as essential components of responsible AI deployment in schools. Ethical implementation becomes particularly salient when educators decide how and when to integrate generative AI into classroom practice. For example, using AI-generated feedback without transparency may obscure the source of evaluation and limit students' ability to question or interpret the guidance they receive. An ethically informed approach would involve clearly communicating how AI tools are used, what data they rely on, and where human judgment remains central.

Ultimately, these four conceptual foundations and their corresponding applied domains form a two-tiered model linking theory to practice. By integrating epistemic, structural, ideological, and policy-oriented perspectives documented across recent scholarship, the IDEA Framework emphasizes that equitable educational AI depends on sustained attention to participation, representation, governance, and institutional responsibility throughout both design and implementation processes.

Discussion

This study examined how AI tools and policies in K-12 education mediate equity, identity, and justice through integrative synthesis of contemporary scholarship informed by critical social theories. Rather than presenting empirical findings, the analysis illuminated how educational AI systems operate as sociotechnical infrastructures that reflect and reproduce broader configurations of power. Across the literature, inequities associated with educational AI emerge not only from biased training data or flawed algorithmic design but also through classroom deployment, institutional policy adoption, and the exclusion of educators and communities from decision-making processes (Delcker, 2024; Gaskins, 2022; Muldoon, 2023). These patterns extend prior critiques of algorithmic oppression by demonstrating how ostensibly neutral educational technologies can reinforce racialized tracking, misclassify student ability, and marginalize culturally situated ways of knowing within K-12 environments.

These findings directly inform the study's guiding question by revealing how AI systems can both reflect and reproduce structural inequities through their design, data, and implementation. The recurring patterns of algorithmic opacity, epistemic exclusion, and governance gaps highlight the need for approaches that move beyond identifying bias toward

actively guiding equitable design and use. In response, the IDEA Framework emerges as a synthesis-driven model that translates these concerns into actionable domains for evaluating and shaping AI integration in K-12 education.

The IDEA Framework was developed in response to these synthesized concerns, translating critical insights into a design-oriented analytic structure for more equitable AI integration. The framework organizes four conceptual foundations—Epistemic and Social Justice, Structural and Design Injustice, Ideological and Historical Foundations, and Policy and Ethical Oversight—into four applied domains: Inclusive Design, Structural Equity, Bias and Justice Analysis, and Ethical Implementation. Rather than offering prescriptive solutions, these domains function as structured entry points through which educators, designers, and policymakers can interrogate how AI systems shape learning experiences, institutional practices, and distributions of authority.

The Inclusive Design domain reinforces and extends contemporary scholarship calling for greater participation and representation in AI development. While recent studies emphasize demographic diversity and inclusive datasets, the synthesized literature indicates that equity-oriented design must also attend to epistemic diversity, community voice, and lived experience throughout iterative development processes (Kamran, 2024; Mohamed, et al., 2020). By foregrounding inclusion as relational and participatory, the IDEA Framework reframes AI development as an educational and ethical practice rather than a purely technical endeavor.

The Structural Equity domain expands dominant equity narratives that focus primarily on device access or infrastructural provision. Although access remains necessary, the synthesis reveals that inequity is frequently embedded within system architecture itself, through opaque

classification logics, standardized benchmarks, proxy variables, and unexamined behavioral assumptions (Dwivedi, 2023; Kohnke, 2024). By centering structural conditions rather than surface-level access, this domain emphasizes the need to interrogate how algorithmic systems distribute opportunity and constraint within educational environments.

The Bias and Justice Analysis domain responds to concerns regarding the normalization of algorithmic decision-making. Contemporary research cautions that AI bias reflects structural patterns shaped by training data, institutional priorities, and historical inequities (Angwin et al., 2016; Selbst et al., 2019). When examined through an intersectional lens, these dynamics reveal how students situated at the intersections of race, gender, class, language, and ability are more likely to experience compounded forms of misclassification and exclusion (Dunne, 2023; Van Slyke, 2025). The framework therefore positions bias analysis as an ongoing critical practice embedded within continuous evaluation rather than a one-time compliance measure.

Finally, the Ethical Implementation domain highlights the role of governance and institutional responsibility in shaping AI adoption. While global and national policy initiatives increasingly emphasize transparency and accountability (UNESCO, 2022; U.S. Department of Education, 2023), the literature suggests that ethical AI integration cannot be reduced to regulatory compliance alone. Justice-oriented oversight requires sustained attention to context, power, participation, and professional judgment, particularly in environments where AI systems may normalize surveillance or automate disciplinary practices (Eutsler et al., 2025; Van Slyke, 2025).

These domains reinforce the central insight that educational AI systems function as value-laden infrastructures embedded within historical, ideological, and institutional contexts.

The IDEA Framework contributes a structured analytic language for examining how AI systems redistribute epistemic authority, structure opportunity, and shape institutional power within K-12 education. By linking critical theory to design-oriented practice, the framework advances ongoing conversations about AI literacy and institutional responsibility (Delcker, 2024; Fullan et al., 2023), affirming that equity in education requires deliberate, justice-centered engagement with technology rather than reliance on automation alone.

Implications and Future Research

This study offers several implications for educators, preservice teacher preparation programs, and institutional leaders seeking to engage with AI tools in more equitable and justice-centered ways. First, the synthesis underscores the importance of integrating ethically informed AI literacy into teacher education curricula. Preservice teachers require opportunities not only to learn how to use AI tools, but also to interrogate the assumptions embedded within them, critically evaluate their outputs, and recognize the social, cultural, and institutional forces shaping their design. This emphasis aligns with calls for critical consciousness in educational practice (Freire, 1970) and reflects emerging scholarship advocating for the inclusion of critical AI literacy within educator preparation programs (Delcker, 2024; Dwivedi, 2023). Preparing future educators to navigate AI systems as reflective professionals rather than passive adopters is essential to sustaining equitable integration.

For practicing K-12 educators, the synthesis highlights the need for professional learning experiences that extend beyond technical training. Effective professional development should foreground inclusive design practices, critical bias analysis, and attention to representational and structural inequities embedded in AI systems. This includes community-informed tool selection,

examination of data representation gaps, and reflection on how AI-mediated processes may affect students differently across language, race, ability, gender, and socioeconomic contexts (Kamran, 2024; Mohamed, et al., 2020). In this context, the IDEA Framework offers a structured analytic lens through which school leaders and educators can examine AI procurement, adoption, and classroom implementation through the domains of Inclusive Design, Structural Equity, Bias and Justice Analysis, and Ethical Implementation, particularly in districts serving historically marginalized populations.

The framework also suggests several directions for future research. Empirical studies situated in classroom contexts are needed to better understand how AI tools are enacted in everyday instructional practice and how algorithmic bias may manifest in teacher decision-making and student experience. Longitudinal research could examine how sustained AI adoption shapes learning environments over time, particularly across intersecting dimensions of race, class, language, gender, and ability. As prior scholarship demonstrates, technological interventions frequently replicate existing inequities when their design and implementation are not critically examined (Eubanks, 2018; Noble, 2018). Investigating these dynamics will be essential for assessing the real-world implications of AI integration.

Participatory design research represents another promising avenue. Studies that position students, families, and educators as co-researchers can disrupt top-down development models and redistribute design authority. Such approaches align with the IDEA Framework's emphasis on epistemic justice and inclusive participation by ensuring that those most affected by algorithmic systems have meaningful opportunities to shape their development and use. This direction builds on equity-oriented scholarship that challenges technocratic models of

educational innovation and advocates for more democratic and community-responsive design practices (Benjamin, 2019).

Finally, additional investigation is needed into AI policy implementation at district, state, and federal levels. Research should examine the effectiveness of transparency mandates, data privacy protections, procurement guidelines, and mechanisms for ongoing bias monitoring. Recent policy analyses underscore the limited scope and enforceability of current oversight structures, highlighting the need for governance frameworks that prioritize accountability, public trust, and educational equity (UNESCO, 2022; U.S. Department of Education, 2023). As AI technologies continue to shape educational practice, future scholarship must remain attentive not only to technological capability, but to the values encoded within these systems and the educational futures they help construct.

Limitations

While this study offers a theoretically grounded integrative synthesis of scholarship on AI and equity in K-12 education, several limitations warrant consideration. First, the analysis is based on a curated corpus of interdisciplinary sources rather than primary empirical data from classroom settings. While the corpus was intentionally curated to support conceptual depth and theoretical alignment, this purposive selection process may introduce potential selection bias by privileging certain perspectives or areas of emphasis within the literature. As such, the findings are not intended to represent an exhaustive account of all educational AI research, but rather a synthesis of key themes relevant to equity-focused analysis and framework development. Although this approach enables conceptual depth and cross-disciplinary integration, it limits the

ability to speak directly to how AI tools are enacted within specific schools, districts, or instructional contexts.

Second, as an interpretive integrative review, the synthesis necessarily reflects analytic judgments made during the selection, organization, and conceptual framing of the literature. The identification of domains and the development of the IDEA Framework were shaped by the study's critical and justice-centered orientation. While this perspective provides coherence and theoretical grounding, it also foregrounds particular equity concerns and may not capture alternative framings present in other strands of the literature.

Additionally, the rapidly evolving nature of educational AI presents inherent limitations. Emerging technologies, localized policy initiatives, and classroom-level implementation practices may not yet be fully represented within the available scholarly record. Consequently, the framework reflects patterns identifiable in contemporary research rather than the complete range of practices unfolding across diverse educational settings.

Finally, the IDEA Framework is conceptual and design-oriented and has not yet been empirically examined in applied contexts. Its practical utility, adaptability, and influence on decision-making remain to be tested through future empirical and design-based research. Systematic investigation of how educators and institutions engage with the framework will be essential for evaluating its capacity to guide equitable AI integration across varied educational environments.

Conclusion

This study examined how AI tools and policies in K-12 education reinforce or challenge systemic inequities when viewed through critical social and educational perspectives. Through integrative synthesis of contemporary scholarship on equity, ethics, and educational AI, the study developed the Inclusive Design for Equitable AI in Education (IDEA) Framework as a conceptual and design-oriented contribution to the field. The framework articulates four interrelated applied domains—Inclusive Design, Structural Equity, Bias and Justice Analysis, and Ethical Implementation—that together provide a structured approach for examining how AI systems shape educational experiences, institutional practices, and distributions of opportunity.

Across the literature, educational AI systems are shown to reflect embedded power dynamics, marginalize non-dominant knowledge systems, and operate within policy environments that often lack transparency and accountability. The IDEA Framework responds to these patterns by foregrounding participation, representation, structural awareness, and ethical responsibility as central considerations in AI design and implementation. The framework reinforces that educational AI systems are value-laden infrastructures requiring sustained critical engagement.

As AI technologies expand within educational settings, integrative frameworks connecting theory, governance, and classroom practice become essential. The IDEA Framework offers one such structure, linking critical analysis to design-oriented reflection in ways that support educators, leaders, and policymakers in making more informed and equitable decisions. Advancing just educational futures in an era of algorithmic mediation will depend not only on technological innovation, but on deliberate commitments to inclusion, accountability, and shared authority in the systems that shape learning. As generative AI continues to expand its presence in K-12 education, attending to issues of equity, agency, and justice is not optional, but essential to

ensuring that these technologies contribute to more just and inclusive learning environments rather than reinforcing existing disparities.

References

- Adams, R. (2021). Can artificial intelligence be decolonized? *Interdisciplinary Science Reviews*, 46(1–2), 176–197. <https://doi.org/10.1080/03080188.2020.1840225>
- Akgun, S., & Greenhow, C. (2021). Artificial intelligence in education: Addressing ethical challenges in K-12 settings. *AI And Ethics*, 2(3), 431–440. <https://doi.org/10.1007/s43681-021-00096-7>
- Benjamin, R. (2019). *Race after technology: Abolitionist tools for the new Jim Code*. Polity Press.
- Braaten, E., Farnsworth, K., The William and Ida Friday Institute for Educational Innovation, & North Carolina State University. (2024). Educators' perspectives on generative AI in K-12: Informing AI in education guidance. <https://fi.ncsu.edu/wp-content/uploads/sites/175/2024/05/Educators-Perspectives-on-Gen-AI-in-K-12V2.pdf>
- Center on Reinventing Public Education. (2023). AI is coming to U.S. classrooms, but who will benefit? <https://crpe.org/ai-is-coming-to-u-s-classrooms-but-who-will-benefit/>
- Bulut, O., & Beiting-Parrish, M. (2024b). The rise of artificial intelligence in educational measurement: opportunities and ethical challenges. *Chinese/English Journal of Educational Measurement and Evaluation*, 5(3). <https://doi.org/10.59863/miq17785>
- Cave, S., & Dihal, K. (2020). The whiteness of AI. *Philosophy & Technology*, 33(4), 685–703. <https://doi.org/10.1007/s13347-020-00415-6>

- Crenshaw, K. (1989). Demarginalizing the intersection of race and sex. University of Chicago Legal Forum, 1989(1), 139–167.
- Delcker, J., Heil, J., & Ifenthaler, D. (2024). Evidence-based development of an instrument for the assessment of teachers' self-perceptions of their artificial intelligence competence. *Educational Technology Research and Development*. <https://doi.org/10.1007/s11423-024-10418-1>
- Dunne, G. (2022). Epistemic injustice in education. *Educational Philosophy and Theory*, 55(3), 285–289. <https://doi.org/10.1080/00131857.2022.2139238>
- Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, A. S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., . . . Wright, R. (2023). Opinion Paper: “So what if ChatGPT wrote it?” Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*, 71, 102642. <https://doi.org/10.1016/j.ijinfomgt.2023.102642>
- EU Artificial Intelligence Act | Up-to-date developments and analyses of the EU AI Act.* (n.d.). <https://artificialintelligenceact.eu/>
- Eubanks, V. (2018). *Automating inequality: How high-tech tools profile, police, and punish the poor*. St. Martin's Press.
- Eutsler, L., Rivera, B., Barnes, M., & Cummings, J. (2025). Artificial intelligence policies in K-12 school districts in the United States: A content analysis shaping education policy.

Journal of Research on Technology in Education, 1-25.

<https://doi.org/10.1080/15391523.2025.2476589>

Farag, A., Greeley, L., & Swindell, A. (2021). Freire 2.0: Pedagogy of the digitally oppressed.

Educational Philosophy and Theory, 54(13), 2214–2227.

<https://doi.org/10.1080/00131857.2021.2010541>

Freire, P. (1970). *Pedagogy of the Oppressed*. Herder and Herder.

Fricker, M. (2007). *Epistemic injustice: Power and the ethics of knowing*. Oxford University Press.

Gaskins, N. (2022). Interrogating algorithmic bias: From speculative fiction to liberatory design.

TechTrends, 67(3), 417–425. <https://doi.org/10.1007/s11528-022-00783-0>

Gouseti, A., James, F., Fallin, L., & Burden, K. (2024). The ethics of using AI in K-12 education: a systematic literature review. *Technology Pedagogy and Education*, 1–22.

<https://doi.org/10.1080/1475939x.2024.2428601>

Gross, N. (2023). What ChatGPT tells us about gender: a Cautionary tale about performativity and gender biases in AI. *Social Sciences*, 12(8), 435.

<https://doi.org/10.3390/socsci12080435>

Holmes, W., & Porayska-Pomsta, K. (2022). *The ethics of artificial intelligence in education*.

Routledge eBooks. <https://doi.org/10.4324/9780429329067>

- Hutchinson, B., Prabhakaran, V., Denton, E., Webster, K., Zhong, Y., & Denuyl, S. (2020). Social biases in NLP models as barriers for persons with disabilities. *Association for Computational Linguistics*. <https://doi.org/10.18653/v1/2020.acl-main.487>
- Karasinski, M., & Candioto, K. B. B. (2024). AI's black box and the supremacy of standards. *Filosofia Unisinos*, 25(1), 1–13. <https://doi.org/10.4013/fsu.2024.251.13>
- Kamran, A. (2023). Decolonizing artificial intelligence: Unveiling biases, power dynamics, and colonial continuities in AI systems. *RMS journal*. <https://ssrn.com/abstract=4610643>
- Kay, J., Kasirzadeh, A., & Mohamed, S. (2024). Epistemic injustice in generative AI. *Cornell University*. <https://doi.org/10.48550/arxiv.2408.11441>
- Kidd, I. J., Spencer, L., & Carel, H. (2022). Epistemic injustice in psychiatric research and practice. *Philosophical Psychology*, 1–29. <https://doi.org/10.1080/09515089.2022.2156333>
- Kohnke, L., & Fount, D. (2024). Deconstructing the normalization of data colonialism in educational technology. *Education Sciences*, 14(1), 57. <https://doi.org/10.3390/educsci14010057>
- Long, D., & Magerko, B. (2020). What is AI Literacy? Competencies and Design Considerations. *Association for Computing Machinery*, 1–16. <https://doi.org/10.1145/3313831.3376727>
- Ma, M., Ng, D. T. K., Liu, Z., & Wong, G. K. (2025). Fostering Responsible AI literacy: A

- Systematic Review of K-12 AI Ethics education. *Computers and Education Artificial Intelligence*, 100422. <https://doi.org/10.1016/j.caeai.2025.100422>
- Medina, J. (2013). *The epistemology of resistance: Gender and racial oppression, epistemic injustice, and resistant imaginations*. Oxford University Press.
- Mohamed, S., Png, M., & Isaac, W. (2020). Decolonial AI: Decolonial theory as sociotechnical foresight in artificial intelligence. *Philosophy & Technology*, 33(4), 659–684. <https://doi.org/10.1007/s13347-020-00405-8>
- Muldoon, J., & Wu, B. A. (2023). Artificial intelligence in the colonial matrix of power. *Philosophy & Technology*, 36(4). <https://doi.org/10.1007/s13347-023-00687-8>
- Nazaretsky, T., Cukurova, M., & Alexandron, G. (2022). An instrument for measuring teachers' trust in AI-Based educational technology. *Association for Computing Machinery*, 56–66. <https://doi.org/10.1145/3506860.3506866>
- Noble, S. U. (2018). *Algorithms of oppression: How search engines reinforce racism*. NYU Press.
- Rice, C., Harrison, E., & Friedman, M. (2019). Doing justice to intersectionality in research. *Culture Studies & Critical Methodologies*, 19(6), 409–420. <https://doi.org/10.1177/1532708619829779>
- Sardelli, M. (2022). *Epistemic Injustice in the Age of AI*. <https://ojs.st-andrews.ac.uk/index.php/aporia/article/view/2455>

Schwartz, R., Vassilev, A., Greene, K., Perine, L., Burt, A., & Hall, P. (2022). *Towards a standard for identifying and managing bias in artificial intelligence*.

<https://doi.org/10.6028/nist.sp.1270>

Southern Regional Education Board (SREB). (2025). A roadmap for responsible and effective use of AI in K-12 classrooms. <https://sreb.org>

UNESCO. (2022). *Recommendation on the ethics of artificial intelligence*.

<https://unesdoc.unesco.org/ark:/48223/pf0000381137>

U.S. Department of Education. (2023). *Artificial intelligence and the future of teaching and learning*. <https://www.ed.gov/ai>

Van Slyke, C.; Sarabadani, J.; and Mosafer, H. (2025). Epistemic injustice in generative AI.

AMCIS 2025 Proceedings. 6. <https://aisel.aisnet.org/amcis2025/intelfuture/intelfuture/6>

Appendix A

Analytic Corpus and Thematic Contributions (2020–2025)

Thematic category	Source	Primary analytic focus
Structural Bias, Epistemic Exclusion, and Design Injustice	Cave, S., & Dihal, K. (2020)	Diversity of AI; standardization bias; epistemic exclusion in system design
	Kay, M. (2024)	Epistemic representation; legitimacy of student knowledge in educational AI systems
	Mohamed, S., et al. (2020)	Decolonial AI; structural bias; epistemic authority; design injustice
Algorithmic Opacity, Assessment, and Surveillance	Bulut, O., & Beiting-Parrish, M. (2024)	AI-driven educational measurement; standardization and assessment bias
	Nazaretsky, T. (2022)	Algorithmic opacity; explainability; teacher trust in AI-mediated decision-making
	Sardelli, A. (2022)	Linguistic bias in automated assessment and scoring systems
	Van Slyke, J. (2025)	AI-enabled student monitoring; surveillance in classroom contexts
Intersectional Harm and Classroom-Level Inequities	Dunne, G. (2022)	Intersectional epistemic harm; classroom-level

		misrecognition of student knowledge
	Gaskins, N. (2022)	Classroom surveillance; algorithmic harm affecting marginalized learners
Governance, Ethics, and Lifecycle Oversight	Center on Reinventing Public Education. (2023)	Structural inequities in AI adoption across U.S. school systems
	Dwivedi, Y. K., et al. (2023)	AI governance; ethics; accountability across the AI lifecycle
	European Union. (2024)	Risk-based regulatory oversight (AI Act)
	Eutsler, L., et al. (2025)	District-level AI policy development processes
	Kohnke, L. (2024)	Educator-oriented AI evaluation and governance tools
	Muldoon, M. (2023)	Extractive data practices; ethical risk in AI systems
	UNESCO. (2022)	Rights-based, human-centered AI governance
	U.S. Department of Education. (2023)	Federal AI policy framing for education
Teacher Agency, Participatory, and Inclusive Design	Delcker, J., Heil, J., & Ifenthaler, D. (2024)	Teacher agency in classroom AI implementation
	Kamran, F. (2024)	Inclusive design; representational harm in AI systems

Schwartz, R. (2022)

Participatory design;
community and educator
voice in technology design

Note. Each source is categorized according to its primary analytic contribution to the integrative synthesis. While several works inform multiple equity dimensions, sources are listed once to ensure conceptual clarity and methodological transparency.